



SentryTheta AI

Autonomous Options Strategy & Institutional Risk Guardrails Desk

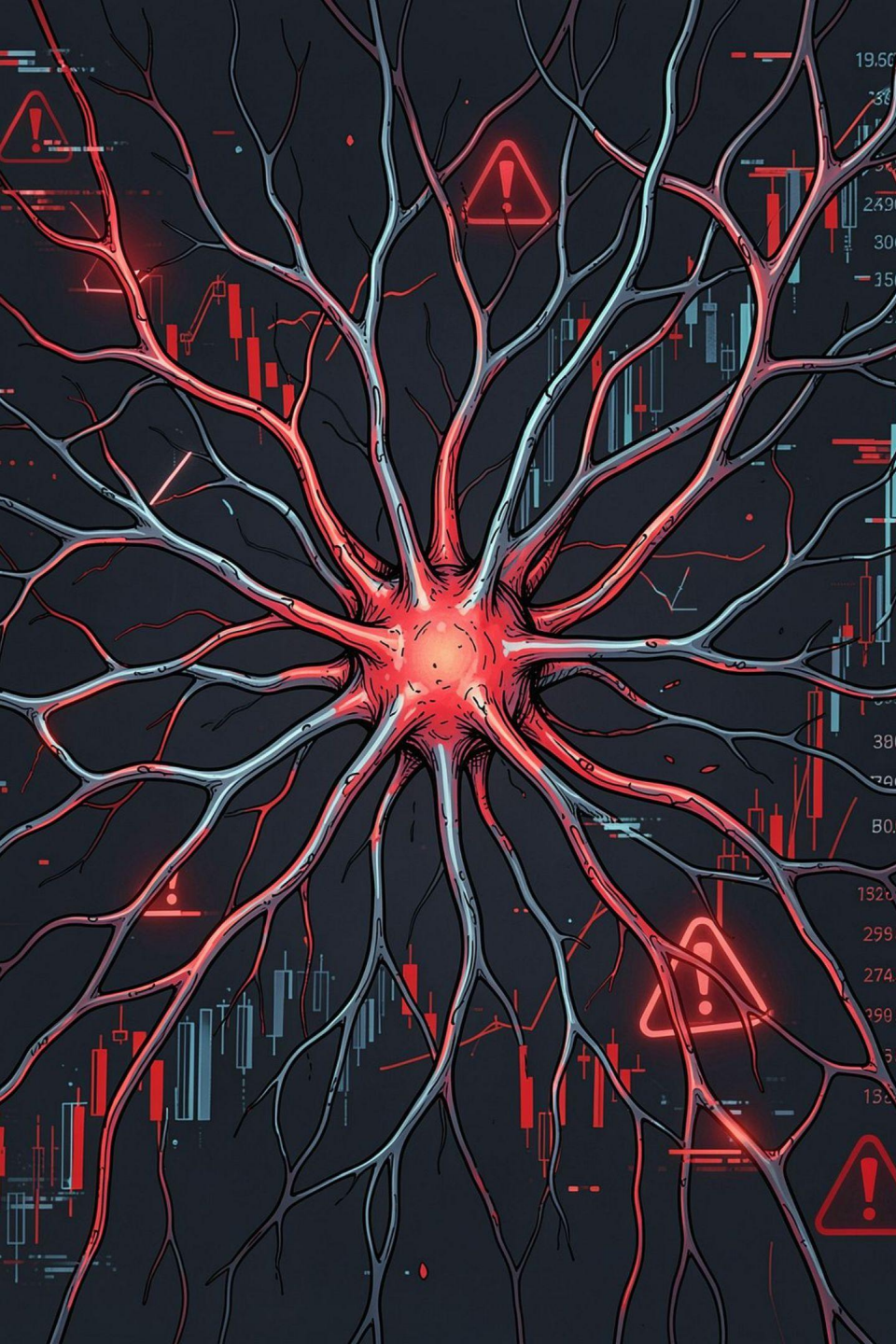
Event: Built for Alpaca AI Trading Agents Hackathon (lablab.ai × Alpaca)

Team:

1st Participant: Muhammad Abdullah

2nd Participant: Agha Essa Khan





The Problem: Why Generative AI Fails in Financial Markets

✗ Unfiltered LLM Hallucinations

Reckless orders without calculating margin or liquidity.

✗ Absence of Capital Preservation

Naive bots over-leverage until portfolio capital is wiped out.

✗ Options Complexity

Strict adherence to Greeks (Theta, Delta, IV) cannot be computed deterministically by raw LLMs.

The Solution & Core Architecture

Separation of Concerns

Decoupled single-responsibility agents for News, Option Chains, Risk, and Execution.

Deterministic Risk Sentry

Hard mathematical gatekeeper that overrides hallucinations.

Orchestrated Pipeline

News -> Analyst AI -> Volatility Strategist -> Risk Audit -> Alpaca Broker Execution.

News

Analyst AI

Volatility Strategist

Risk Audit



Agent 1: Market Analyst AI



📡 Automated RSS Feed Scanning

Parses real-time Yahoo Finance RSS news feeds for target tickers.



🧠 LLM-Powered Sentiment Analysis

Uses Google Gemini 1.5/2.0 Flash to calculate normalized sentiment scores (-1.0 to +1.0).



🔍 Catalyst Extraction

Filters raw market noise into actionable directional signals.

Agent 2: Volatility Strategist

Live Option Chain Retrieval

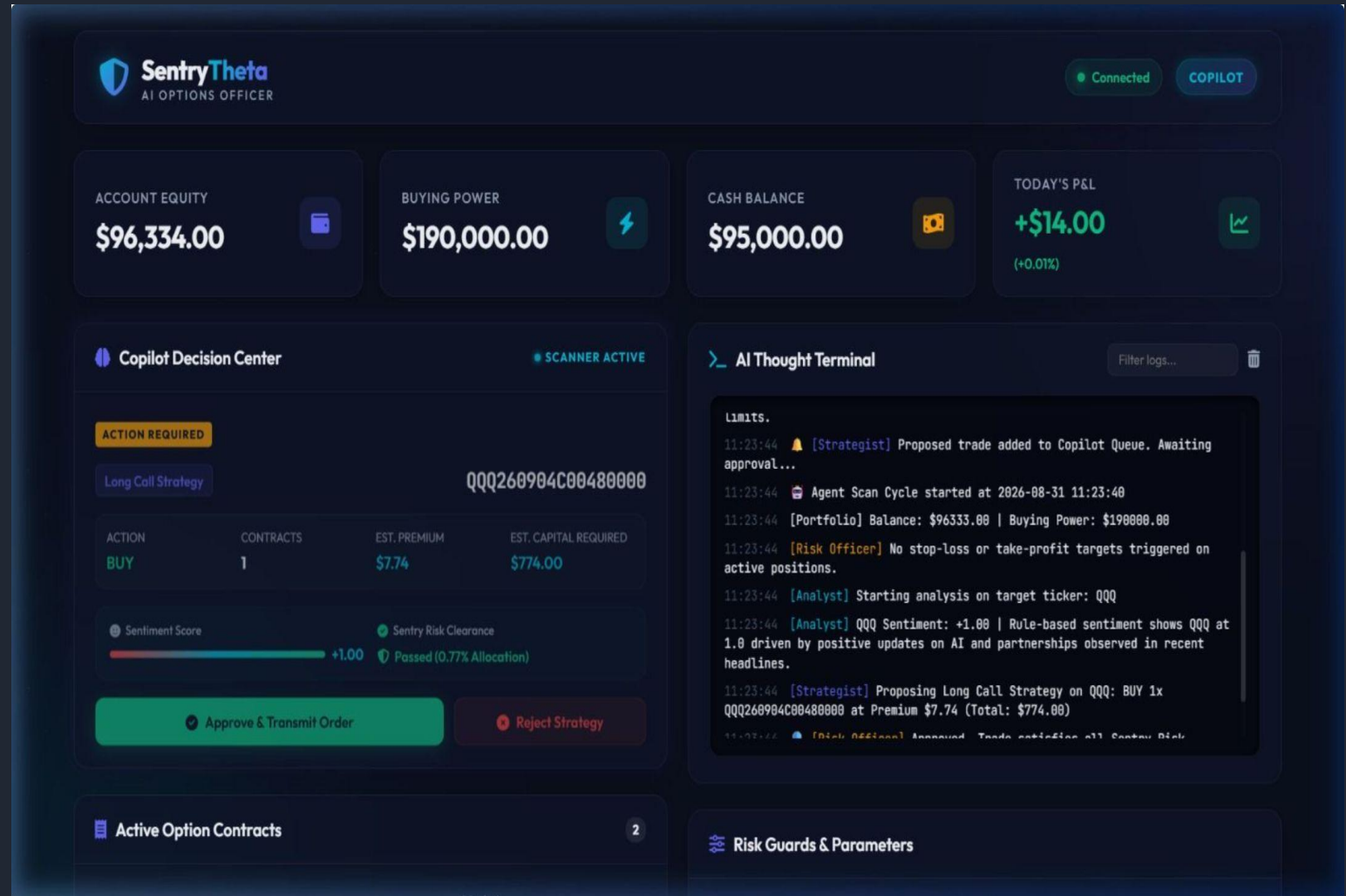
Queries Alpaca option chains for active contract strikes, expiries, and liquidity.

Defined-Risk Options Structuring

Formulates Long Calls, Long Puts, and Cash-Secured Puts matching sentiment polarity.

Expiry Optimization

Selects near-the-money options with optimal expiry dates.



The screenshot displays the SentryTheta AI Options Officer dashboard. At the top, the user is logged in as 'Connected' and has access to 'COPILOT'. The dashboard features four main summary cards: ACCOUNT EQUITY (\$96,334.00), BUYING POWER (\$190,000.00), CASH BALANCE (\$95,000.00), and TODAY'S P&L (+\$14.00, +0.01%). Below these is the 'Copilot Decision Center' with a 'SCANNER ACTIVE' status. It shows an 'ACTION REQUIRED' for a 'Long Call Strategy' on the ticker 'QQQ260904C00480000'. A table details the proposed trade: BUY 1 contract at an estimated premium of \$7.74, requiring \$774.00 in capital. Below the table, a 'Sentiment Score' of +1.00 is shown, along with 'Sentry Risk Clearance' status: 'Passed (0.77% Allocation)'. Two buttons are available: 'Approve & Transmit Order' and 'Reject Strategy'. To the right is the 'AI Thought Terminal' showing a log of events: a proposed trade added to the queue, an agent scan cycle starting, portfolio balance and buying power updates, a risk officer's confirmation of no stop-loss or take-profit targets, an analyst starting analysis on the target ticker, sentiment analysis results (QQQ Sentiment: +1.00), and a final proposal for a long call strategy. At the bottom, there are sections for 'Active Option Contracts' (2) and 'Risk Guards & Parameters'.

SentryTheta
AI OPTIONS OFFICER

Connected COPILOT

ACCOUNT EQUITY
\$96,334.00

BUYING POWER
\$190,000.00

CASH BALANCE
\$95,000.00

TODAY'S P&L
+\$14.00
(+0.01%)

Copilot Decision Center SCANNER ACTIVE

ACTION REQUIRED

Long Call Strategy **QQQ260904C00480000**

ACTION	CONTRACTS	EST. PREMIUM	EST. CAPITAL REQUIRED
BUY	1	\$7.74	\$774.00

Sentiment Score: +1.00
Sentry Risk Clearance: Passed (0.77% Allocation)

Approve & Transmit Order Reject Strategy

AI Thought Terminal Filter logs...

LIMITS.

11:23:44 [Strategist] Proposed trade added to Copilot Queue. Awaiting approval...

11:23:44 Agent Scan Cycle started at 2026-08-31 11:23:40

11:23:44 [Portfolio] Balance: \$96333.00 | Buying Power: \$190000.00

11:23:44 [Risk Officer] No stop-loss or take-profit targets triggered on active positions.

11:23:44 [Analyst] Starting analysis on target ticker: QQQ

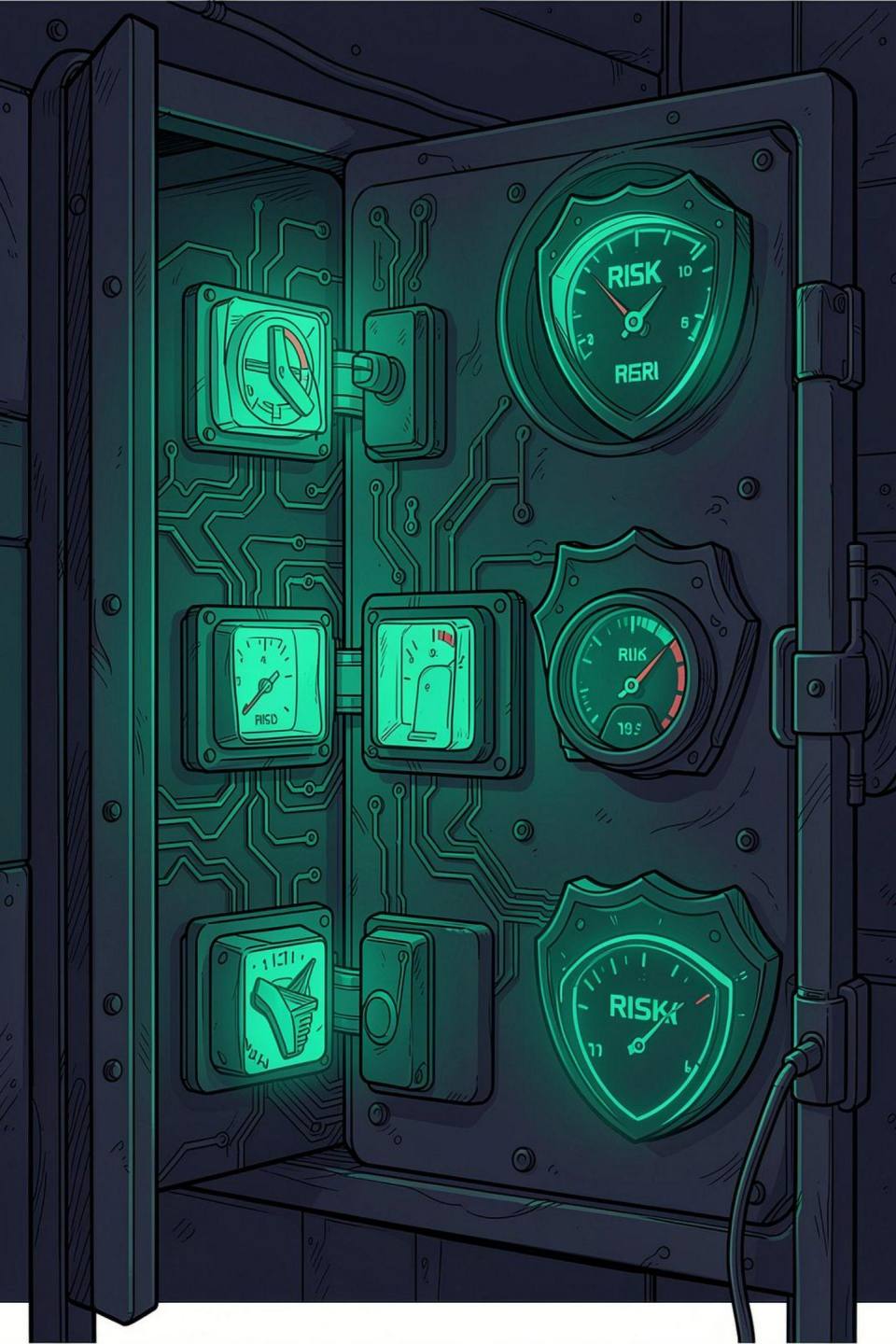
11:23:44 [Analyst] QQQ Sentiment: +1.00 | Rule-based sentiment shows QQQ at 1.0 driven by positive updates on AI and partnerships observed in recent headlines.

11:23:44 [Strategist] Proposing Long Call Strategy on QQQ: BUY 1x QQQ260904C00480000 at Premium \$7.74 (Total: \$774.00)

11:23:44 [Risk Officer] Approved Trade satisfies all Sentry Risk

Active Option Contracts 2

Risk Guards & Parameters



Agent 3: Sentry Risk Officer (Risk Engine)



Position Sizing Cap (5% Equity)

Limits single option premium cost to max 5% total account equity.



Daily Drawdown Circuit Breaker (3% Limit)

Instantly halts trading if daily portfolio drawdown reaches 3%, protecting 97%+ capital.



Exit Harvesting

Auto-liquidates active contracts at Stop-Loss (-15%) or Take-Profit (+30%).

Execution Modes & Glassmorphic UI Dashboard

Dual Execution Modes

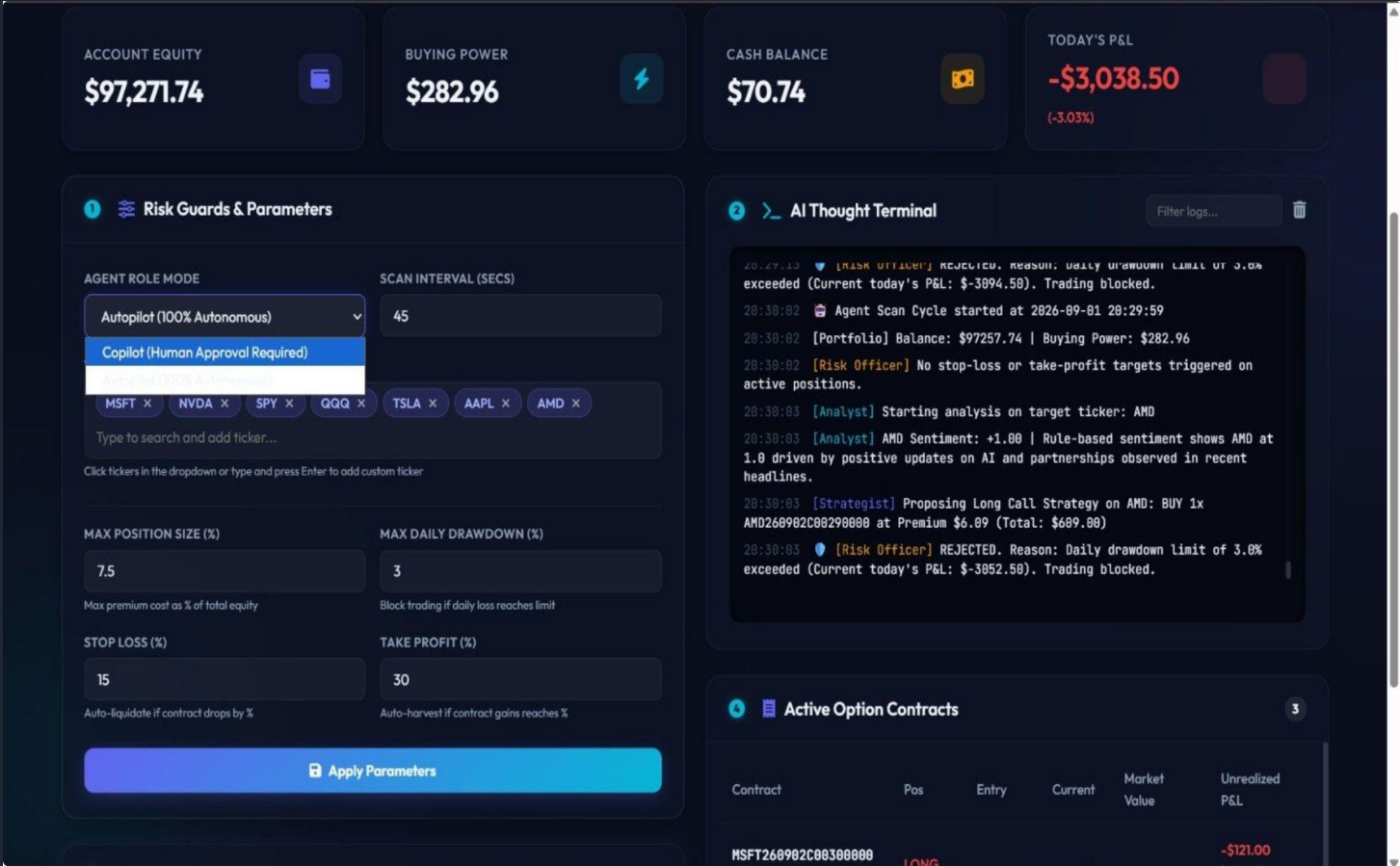
Copilot (1-click human review) vs.
Autopilot (100% autonomous live paper execution).

4-Step Guided Layout

[1] Configure Limits → [2] AI Scan Loop →
[3] Approve Order → [4] Monitor Performance.

AI Thought Terminal

Live color-coded scrolling logs for each agent.



Technology Stack & Infrastructure



⚙️ Backend & AI

Python 3.11, FastAPI, Google Gemini API, Yahoo Finance RSS, alpaca-py SDK.



💾 State & MCP

SQLite (sentrytheta.db) persistence across restarts, Model Context Protocol (MCP) server integration.



🎨 Frontend & Styling

HTML5, Modern CSS Variables, Glassmorphic UI layout.

Future Roadmap: Scaling SentryTheta AI

1

🌐 Phase 1 – Multi-Broker Expansion

Connectors for Tradier, Interactive Brokers, and Charles Schwab.

2

📊 Phase 2 – Multi-Leg Options

Advanced delta-neutral spreads (Iron Condors, Strangles, Butterfly Spreads).

3

🤖 Phase 3 – Macro & Vision AI

Multi-modal chart pattern recognition & macroeconomic fed rate hedging.



Conclusion & Q&A

Intelligent Generative Reasoning Supervised by Unbreakable Quantitative Risk.

Thanks to lablab.ai and Alpaca for hosting the Alpaca AI Trading Agents Hackathon.



Open for Questions & Feedback!

